

Morning Paper Report - Friday, July 31, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Transformed Visual Working Memory Representations in Human Occipitotemporal and Posterior Parietal Cortices

Yaoda Xu
eNeuro, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory representation, working memory

Link: <https://www.semanticscholar.org/paper/c4bad8e9cbf77ac0391427c3b71f7f1c78f17733>

Goal: Recent fMRI studies reported transformed representations between perception and visual working memory (VWM) in the human early visual cortex (EVC)

Method: Although cross-decoding was usually lower than within-VWM decoding and consistent with transformed VWM representations, this has been attributed to experimental differences between perceptual and VWM tasks: once they are equated, the same representation is expected to exist in both

Results: Even with strong VWM representations present throughout the occipitotemporal cortex (OTC, including EVC) and posterior parietal cortex (PPC), fMRI cross-decoding revealed significant representational differences between distractors (perception) and targets (VWM) in both regions

Conclusion: Transformed representations, mediated by task goals and associative areas coding task-relevant information (e.g., PPC), can easily account for these and other recent findings. A task-driven transformed account of VWM thus better captures the nature of VWM representation in the human brain (including EVC) than the sensory representations originally proposed by the sensory account of VWM.

2. Reviewer #1 (Public review): Neural dynamics of visual working memory representation during sensory distraction

Authors unavailable
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory

Link: <https://doi.org/10.7554/elife.99290.4.sa1>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

3. The Impact of Visual Working Memory Chunking on Visual Search

Logan Doyle, Susanne Ferber
Journal of Vision, 2023

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: <https://doi.org/10.1167/jov.23.9.5160>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

4. Can working memory be explained by predictive coding?

Mengli Feng

2023

Source: Crossref; matched terms: working memory, modelling, artificial intelligence, predictive coding

Link: <https://doi.org/10.31237/osf.io/sqfr9>

Goal: Predictive coding (PC) is a theory in cognitive/computational neuroscience which explains cortical functions with a hierarchical process of minimising prediction errors

Method: Based on the reviewed literature, I summarised three important ingredients for modelling WM which are temporal depth, goals and hierarchy

Results: Since its discovery, predictive coding has been found to be a unifying theory explaining more and more cognitive functions, including perception, attention, and action planning

Conclusion: Modelling WM in PC frameworks provides alternative explanations to some long-standing questions about WM and may help with resolving the conflicts between WM theories, for example, the ones that propose either persistent or sparse neuronal activity during WM. It may also help with developing computational tools to improve treatments for brain disorders such as schizophrenia and facilitate artificial intelligence...

5. Predicted Utility Modulates Working Memory Fidelity in the Brain

Emily J. Levin, James A. Brissenden, Alexander Fengler, David Badre

2021

Source: Crossref; matched terms: working memory, fidelity, precision

Link: <https://doi.org/10.1101/2021.04.01.438095>

Goal: Abstract The predicted utility of information stored in working memory (WM) is hypothesized to influence the strategic allocation of WM resources

Method: Using a generative model multivoxel analysis approach to estimate the quality of remembered representations across predicted utility conditions, we observed that items with greater predicted utility are maintained in memory with greater fidelity, even when they are the only item being maintained

Results: Abstract The predicted utility of information stored in working memory (WM) is hypothesized to influence the strategic allocation of WM resources

Conclusion: This evidence supports a theoretical distinction between resource allocation that occurs as a result of load and resource pre-allocation that occurs as a result of predicted utility.